Non-symmetric verbal pre-multicategories

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Abstract

Explore non-symmetric substructure of verbal subcategories.

1 Pre-multicategories and pre-operads

We will use the following manipulations of lists. Given a list of elements $\Gamma = \langle a_1, \dots, a_n \rangle$, we define its **length** as $|\Gamma| \coloneqq n$. We allow length-0 lists. A **position** x in a list Γ is a natural number $1 \leq x \leq |\Gamma|$, and so only lists with length 1 and above have positions in them. If $\Gamma = \langle a_1, \dots, a_n \rangle$ we call a_x the **element at position** x **of** Γ. We write $(x : a) \in \Gamma$ when a is at position x in Γ. If $(x : a) \in \Delta = \langle a_1, \dots, a_n \rangle$ and $\Gamma = \langle b_1, \dots, b_k \rangle$ is any list, we define the **substituted** list:

$$(\textbf{let } x = \Gamma \textbf{ in } \Delta) \coloneqq \left\langle a_1, \dots, a_{x-1}, b_1, \dots, b_k, a_{x+1}, \dots, a_n \right\rangle$$

If $x \in \Gamma$ and $y \in \Delta$ we define the **weakening** of y by x as the position:

$$y\,[x] \coloneqq x - 1 + y \in \textbf{let } x = \Gamma \textbf{ in } \Delta$$

We then have that if $(y : b) \in \Gamma$ then $(y\,[x] : b) \in \textbf{let } x = \Gamma \textbf{ in } \Delta$.

▶ **Example 1.** Let $\Gamma = a_1, a_2, a_3$ and $\Delta = b_1, b_2$, then $(2 : b_2) \in \Delta$, when weakened by $(3 : a_3) \in \Gamma$ gives:

$$2\,[3] = 3 - 1 + 2 = 4 \in \langle a_1, a_2, b_1, b_2 \rangle = \textbf{let } x = \Gamma \textbf{ in } \Delta$$

and indeed b_2 is in position 4 there. ◀

▶ **Definition 1** (Staton and Levy 2013)**.** *A* ***pre-multicategory*** $\mathcal{M}$ *consists of:*

- *a collection of* ***objects*** $a \in \mathcal{M}$*;*
- *for every list of objects* Γ *and object* $b \in \mathcal{M}$*, we have a collection* $\mathcal{M}(\Gamma; b)$ *of* ***multi-arrows*** $f : \Gamma \to b$*; when* $|\Gamma| = 1$ *we call a multi-arrow* $f : \langle a \rangle \to b$ *an* ***arrow****, and write instead* $f : a \to b$*; and*
- *for every object* $a \in \mathcal{M}$ *we have a distinguished arrow* $\mathrm{id}_a : a \to a$*;*
- *for every multi-morphism* $g : \Delta \to c$*, position* $(x : b) \in \Delta$ *(i.e.,* $\Delta = \langle b_1, \dots, b_k \rangle$*,* $1 \leq x \leq k$ *and* $b_x = b$*), and* $f : \Gamma \to b$*, we have the* ***composition*** *multi-arrow:*

$$\textbf{let } x = f \textbf{ in } g : \textbf{let } x = \Gamma \textbf{ in } \Delta$$

such that:

- *identities are neutral in any position:*

$$\frac{(x : a) \in \Gamma \qquad f : \Gamma \to b}{\textbf{let } x = \mathrm{id}_a \textbf{ in } f = f}$$

composing into an identity is neutral:

$$\frac{f : \Gamma \to b}{\textbf{let } 1 = f \textbf{ in } \mathrm{id}_b = f}$$

composition is associative:

$$\frac{f : \Gamma \to b \qquad (x : b) \in \Delta \qquad g : \Delta \to c \qquad (y : c) \in \Xi \qquad h : \Xi \to d}{\textbf{let } y = (\textbf{let } x = f \textbf{ in } g) \textbf{ in } h = \textbf{let } x\,[y] = f \textbf{ in let } y = g \textbf{ in } h}$$

OK: TODO: some examples.

OK: TODO: pre-operad as a 1-object pre-multicategory.

OK: TODO: define pre-operadic algebra in a pre-multicategory.

OK: TODO: define verbal pre-multicategories and verbal pre-operads.

OK: TODO: define verbal pre-operadic algebra in a pre-multicategory.

OK: TODO: the notion of a pre-operadic algebra in the pre-multicategory of sets and products is the same as a operadic algebra in the multicategory of sets and products.

OK: TODO: define verbal pre-operadic monad

OK: TODO: show that pre-operadic and operadic monads coincide

OK: TODO: recall verbally commutative monad

OK: TODO: prove the theorem they should´ve proven from the WG meeting.

OK: TODO: if $\boldsymbol{S}$ is $\mathbb{W}$-commutative, then its Kleisli pre-multicategory is a $\mathbb{W}$-pre-multicategory

OK: TODO: I think by this point we have a distributive law between free monads and any monad, by taking $\mathbb{W} = \mathbb{F}_{\mathrm{id}}$.

OK: TODO: consider whether to start on Yoneda structures and distributive combinations

References

Sam Staton and Paul Blain Levy. Universal properties of impure programming languages. In *Proceedings of the 40th Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, POPL '13, page 179–192, New York, NY, USA, 2013. Association for Computing Machinery. ISBN 9781450318327. 10.1145/2429069.2429091. URL `https://doi.org/10.1145/2429069.2429091`.

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